

CHAPTER 16 SUMMARY

Electric Potential Energy

(Section 16.1) A system of two or more electric charges has **potential energy**. Potential energy is solely a function of the position(s) of the particle(s).

The change in potential energy of a particle that moves from one point to another in an electric field is independent of the path taken.

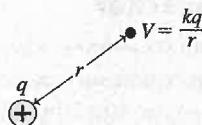
Potential energy of two point charges: $U = \frac{kq_1q_2}{r}$



Electric Potential

(Section 16.2) **Electric potential** is potential energy per unit charge.

Electric potential of a point charge: $V = \frac{kq}{r}$ (with zero of potential taken at an infinite distance)



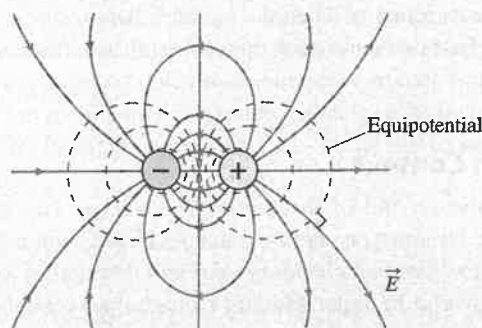
Potential energy and potential: $U = q_0V$

Electric Potential and Electric Field

(Section 16.3) Electric field lines are perpendicular to equipotential surfaces and are directed from higher to lower potential.

An electric field pushes positive charges from higher to lower potential and negative charges from lower to higher potential.

Electric field and potential difference: $E = -\Delta V/\Delta x$ (for uniform electric field)



Capacitors

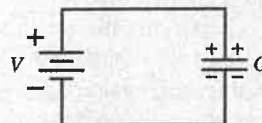
(Section 16.4) A **capacitor** consists of two insulated conductors. Capacitors store charge and energy.

Capacitance is the ratio of charge to potential difference.

Capacitance: $C = \frac{Q}{V}$

Capacitors in parallel: $C_p = C_1 + C_2$

Capacitors in series: $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$



Dielectrics

(Section 16.5) A **dielectric** between capacitor plates increases capacitance and allows the capacitor to store more charge.

Capacitor with a dielectric: $C = \kappa C_0$

NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled B10 are of biological or medical interest.

Conceptual Questions

1. Suppose you have three negative point charges at different locations. Is the electric potential energy greater than or less than zero, or could it be either?
2. Repeat the preceding question for two negative charges and one positive charge.
3. If you have two charges of the same magnitude but opposite sign, separated by some distance d , is there any place where the electric potential is zero? Explain.
4. Repeat the preceding question for the case of two charges that have the same magnitude and same sign.
5. Lightning carries a tremendous amount of electrical energy. Into what forms do you see this energy transformed?
6. If an electron and a proton are each accelerated from rest to a kinetic energy of 300 eV, which one is traveling faster?
7. Is it possible for two equipotential surfaces to intersect? Why or why not?
8. At a point where the electric potential is zero, is the electric field necessarily zero too? Why or why not?
9. At a point where the electric field is zero, is the electric potential necessarily zero too? Why or why not?
10. Two identical positive charges $+Q$ are separated by a distance d . Sketch some electric field lines and equipotentials.
11. Using the electric potential map shown in Figure CQ16.11, indicate the approximate directions and relative magnitudes of the electric field vectors at the points labeled A, B, C, and D.

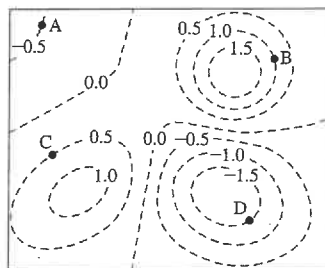


FIGURE CQ16.11

Electric potential (V)

12. Show that the units $C^2/(N \cdot m^2)$ for the constant ϵ_0 reduce to F/m.
13. Suppose you have a parallel-plate capacitor attached to a battery with fixed potential difference V . If you increase the plate spacing, explain whether each of the following increases, decreases, or remains the same: (a) the capacitance; (b) the charge on the plates; (c) the stored energy.
14. Suppose you disconnect the capacitor of the preceding problem from the battery before you move the plates, ensuring that the charge Q is fixed. As you increase the plate separation, explain whether each of the following increases, decreases, or remains the same: (a) the capacitance; (b) the potential difference between the plates; (c) the stored energy.
15. An air-filled parallel-plate capacitor with plate separation d has capacitance C_0 . A dielectric with dielectric constant κ and thickness $d/2$ is inserted between the plates as shown in

Figure CQ16.15. What's the new capacitance? Does it matter whether the dielectric is closer to one plate or the other? Explain.

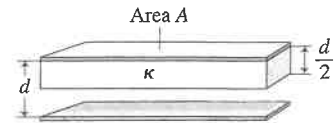


FIGURE CQ16.15

16. An air-filled parallel-plate capacitor with plate area A and separation d has capacitance C_0 . A dielectric with cross-sectional area $A/2$, dielectric constant κ , and thickness d is inserted between the plates as shown in Figure CQ16.16. What's the new capacitance?

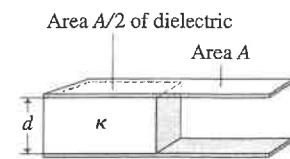


FIGURE CQ16.16

17. An air-filled parallel-plate capacitor is charged to some fixed value Q . A dielectric slab is then inserted between the plates. Is the work done by an external agent to insert the slab positive, negative, or zero? Explain.
18. For the situation with two like charges described in Conceptual Example 16.1, make a plot showing the speed of charge q_2 as a function of its position on the x -axis.

Multiple-Choice Problems

19. Electric potential is (a) force per unit charge; (b) potential energy per unit charge; (c) force per unit distance; (d) electric field per unit charge.
20. Two charges, one positive and one negative, are separated by a fixed distance. The sign of the potential energy of this pair (a) is always positive; (b) is always negative; (c) is sometimes positive and sometimes negative; (d) depends on the distance.
21. A helium nucleus consists of two protons (charge $e = 1.60 \times 10^{-19}$ C) and two neutrons (zero charge). Assuming the protons are separated by 1.9×10^{-15} m, the electric potential energy is (a) 4.8×10^{-13} J; (b) 2.4×10^{-19} J; (c) 3.4×10^{-11} J; (d) 1.2×10^{-13} J.
22. The distance from a proton at which the electric potential is 1 V is (a) 1.4×10^{-9} m; (b) 3.8×10^{-5} m; (c) 2.0×10^{-18} m; (d) 7.2×10^{-5} m.
23. A square 2.5 cm on a side has identical charges -7.1×10^{-9} C on three of its corners. The electric potential at the fourth (empty) corner is (a) 4400 V; (b) -4400 V; (c) -6900 V; (d) -3500 V.
24. When a proton at rest is accelerated through a potential difference of -40 V, its final speed is (a) c (the speed of light); (b) 4000 m/s; (c) 880 m/s; (d) 8.8×10^4 m/s.
25. Two parallel conducting plates carry uniform charge densities $\pm 3.7 \times 10^{-6}$ C/m². If the plate separation is 3.0 mm, the potential difference between the plates is (a) 830 V; (b) 1250 V; (c) 66 V; (d) 130 V.

26. A particle with charge $-3e$ is accelerated from rest through a potential difference of 150 V. This gives the particle a kinetic energy of (a) 150 eV; (b) 50 eV; (c) 450 eV; (d) -150 eV.
27. What's the potential difference across a 380-nF capacitor carrying a charge of $15\text{ }\mu\text{C}$? (a) 200 V; (b) 26 mV; (c) 26 V; (d) 39 V.
28. A $\mu\text{F} \cdot \text{mV}$ is equivalent to a (a) mC; (b) μC ; (c) nC; (d) pC.
29. The equivalent capacitance of 10-F, 20-F, and 30-F capacitors in series is (a) 0.18 F; (b) 5.5 F; (c) 32 F; (d) 60 F.
30. If you want to store 25 J of energy in a capacitor charged to 12 V, the required capacitance is (a) 0.35 F; (b) 2.9 F; (c) 2.1 F; (d) 0.48 F.
31. In terms of the charge stored (Q) and the potential difference (V), the energy stored in a capacitor is (a) $2QV$; (b) QV ; (c) $\frac{1}{2}QV$; (d) Q/V .
32. The capacitance of a water-filled parallel-plate capacitor with plate area $2.4 \times 10^{-3}\text{ m}^2$ and separation 0.065 mm is (a) 520 nF; (b) 0.52 nF; (c) 72 μF ; (d) 26 nF.

Problems

Section 16.1 Electric Potential Energy

33. ■ Two protons in a nucleus are separated by $6.5 \times 10^{-15}\text{ m}$. What's the potential energy of the pair?
34. ■■ In a classical model of the hydrogen atom, the proton and electron are $5.29 \times 10^{-11}\text{ m}$ apart. Compute their *gravitational* potential energy and compare it with the electric potential energy computed in the text. That is, find the ratio of the electric to the gravitational potential energy.
35. ■■ Suppose both Earth and Moon carried the same net charge Q spread uniformly over their surfaces. (a) How large would Q need to be so that the system's electric and gravitational potential energies had equal magnitude? (b) Under these conditions, what would be the average charge per square meter on Earth's surface? (c) Comment on the result of part (b).
36. ■■ Two identical 25-g particles each carry $+1.0\text{ }\mu\text{C}$ of charge. One is held fixed, and the other is placed 2.5 cm away and released. (a) Describe the subsequent motion of the second charge. (b) Find the speed of the moving charge when it's (i) 10 cm from the fixed charge and (ii) 1.0 km from the fixed charge.
37. ■■ Find the potential energy of charges q_1 and q_2 located in the x - y plane: $q_1 = -3.6 \times 10^{-9}\text{ C}$ at (0.12 m, 0.45 m) and $q_2 = 1.6 \times 10^{-9}\text{ C}$ at (0.36 m, 0.55 m).
38. ■■■ Find the potential energy of the charges q_1 and q_2 at the following positions: $q_1 = -3.6 \times 10^{-6}\text{ C}$ at (1.1 m, 1.4 m, 2.4 m) and $q_2 = -1.6 \times 10^{-6}\text{ C}$ at (0.85 m, 1.9 m, 1.4 m).
39. ■■ Three point charges of $2.0\text{ }\mu\text{C}$, $4.0\text{ }\mu\text{C}$, and $6.0\text{ }\mu\text{C}$ form an equilateral triangle with each side 2.5 cm. Find the electric potential energy of this distribution.
40. ■■ A proton is held at a fixed position. An electron is placed 0.75 m away and released. Find the electron's speed when it's 1.0 mm from the proton.
41. ■■ Two protons are released from rest 5.0 mm apart. (a) Describe the subsequent motion after they're released. (b) How fast are the protons moving when they're a great distance apart?

Section 16.2 Electric Potential

42. ■ If the potential difference between a cloud and the ground is about 12 MV, how much charge is transferred in a lightning bolt that releases 1.6 GJ of energy?
43. ■ At 1.0 m from a point charge, the electric potential is 280 V. (a) What's the potential 2.0 m from the same charge? (b) What are the magnitude and sign of the charge?

44. ■■ Two point charges, $+2.0\text{ }\mu\text{C}$ and $-4.0\text{ }\mu\text{C}$, lie 0.45 m apart on the x -axis. Are there any points on the x -axis where the electric potential is zero? If so, find the point(s). If not, explain.
45. ■ After walking across a dry carpet, there's a potential difference of 4500 V between your hand and the doorknob. As you reach toward the knob, a spark jumps, releasing energy $1.6\text{ }\mu\text{J}$. (a) How much charge was transferred? (b) Assuming all the charges were electrons, how many were there?
46. ■■ It's generally okay to ignore relativistic effects when a particle's speed is less than 1% of the speed of light. Find the accelerating potential difference that will bring each of the following particles to that speed, starting from rest: (a) electrons; (b) protons; (c) alpha particles (helium nuclei, consisting of two protons and two neutrons).
47. ■■ A 900-kg electric car uses a 270-V battery pack. As the car accelerates from rest, 400 C flows from the battery. What's the maximum speed the car can attain during this time? (The actual speed will be less, due to friction and other losses.)
48. ■■ In a mass spectrometer, ions accelerate through a fixed potential difference of 150 V. Find the kinetic energy and speed of the following ions after they've accelerated from rest: (a) He^{2+} ; (b) $^{107}\text{Ag}^+$; (c) $^{109}\text{Ag}^+$; (d) $^{235}\text{U}^+$.
49. ■■ The "charmed lambda" particle (symbol Λ^+) is a subatomic particle with mass $4.1 \times 10^{-27}\text{ kg}$ and charge $+e$. Find the potential difference needed to accelerate the Λ^+ from rest to $3.0 \times 10^6\text{ m/s}$ (1% the speed of light).
50. ■■ X rays are produced in a process called "Bremsstrahlung," in which electrons slow and convert their kinetic energy into electromagnetic energy. Find the energy of an x ray produced when an electron moving at 2.8 Mm/s is suddenly brought to rest. Express your answer in both J and eV.
51. ■■ (a) An ion that passes through a 300-V potential difference gains $9.6 \times 10^{-17}\text{ J}$ of kinetic energy. What's its charge? (b) Repeat for an ion that loses the same amount of kinetic energy.
52. ■■ An electron moves through an electric field, and its speed drops from 2500 m/s to 1500 m/s. What's the potential difference between the two points at which the speed was measured?

Section 16.3 Electric Potential and Electric Field

53. ■ The potential difference between a pair of parallel conducting plates is 65 V. The plates are 3.2 cm apart. What's the magnitude of the electric field between the plates?
54. ■■ You're in a uniform electric field of magnitude 260 N/C. Compute the potential difference if you travel 1 cm in each of the following directions: (a) in the direction of the field; (b) opposite to the field; (c) perpendicular to the field; (d) at a 45° angle to the field.
55. ■■ A pair of parallel plates produces a uniform electric field of 850 V/m. If an electron is released at rest from the negative plate, how fast is it moving when it hits the positive plate, 7.5 mm away?
56. ■■ A uniform electric field of about 100 V/m, pointing straight down, exists just above Earth's surface. (a) Find the potential difference between the ground and the top of a 75-m-tall building. (b) What uniform surface charge density would be required on Earth's surface to create such a field? What would be the sign of the surface charge?
57. ■ Two parallel plates separated by 6.5 mm have a potential difference of 250 V between them. (a) Find the electric field between the plates. (b) What's the force on an electron in this field?

58. ■■ Consider a 1.5-C point charge, fixed at the origin. (a) Compute the electric potential 5.00 m away on the x -axis. (b) Compute the electric potential a distance 5.01 m away on the x -axis. (c) Compute the electric field a distance 5.00 m away on the x -axis. (d) Using your answer from parts (a)–(c), show that the relationship $E = -\Delta V/\Delta x$ is approximately true.
59. ■ Spherical equipotential surfaces surround a point charge. If the electric potential is -175 V on a sphere of radius 0.65 m, what is the charge?
60. ■■ A uniform electric field exists between two conducting plates separated by 1.5 mm. A dust grain with charge 65 nC between the plates feels a 25-N electric force. Find (a) the electric field and (b) the potential difference between the plates.
61. ■■ In a Millikan oil-drop experiment used to measure the elementary charge (Chapter 15), two horizontal parallel plates are 1.0 cm apart. With a 180-V potential difference between the plates, a charged droplet of mass 8.8×10^{-16} kg is suspended motionless. (a) If the upper plate is at the higher potential, is the charge on the droplet positive or negative? (b) What's the electric field between the plates under these conditions? (c) Compute the charge on the droplet. To how many elementary charges (e) does this correspond?

Section 16.4 Capacitors

62. ■ Suppose you want to make a parallel-plate capacitor with square plates 1 cm on a side. (a) How far apart should the plates be for a capacitance of 3.0 pF? (b) Repeat for $C = 3.0$ μ F. (c) Are both designs practical? Comment.
63. ■ A 250-pF parallel-plate capacitor consists of two identical circular plates 0.0875 mm apart. Find their radius.
64. ■ Find the equivalent capacitance of three capacitors $C_1 = 4.0$ μ F, $C_2 = 9.0$ μ F, and $C_3 = 12$ μ F, when they're (a) in parallel and (b) in series.
65. ■■ A 0.25-F capacitor and a 0.45-F capacitor are in parallel. (a) Find the equivalent capacitance. (b) Find the charge stored on each capacitor and the total charge stored when the parallel combination is connected across a 24-V battery.
66. ■■ Two capacitors, $C_1 = 4.0$ mF and $C_2 = 6.0$ mF, are in series. (a) Find the equivalent capacitance. (b) If the combination is connected across a 16-V battery, find the charge stored on each capacitor and the potential difference across each.
67. ■■■ Suppose the three-capacitor network in Example 16.6 (Figure 16.20a) is connected across a 12-V battery. Follow these steps to find the charge on each capacitor: (a) Use the equivalent capacitance to find the charge supplied by the battery. (b) Note that the charge supplied by the battery equals the charge Q_3 on the capacitor C_3 . (Why?) Use this fact to compute the potential difference V_3 across C_3 . (c) Compute the potential difference across the other two capacitors. (d) Use your answer from part (c) to find the charge stored on each of C_1 and C_2 . Note: Q_1 and Q_2 aren't the same.
68. ■■ For the capacitor network shown in Figure P16.68, (a) find the equivalent capacitance; (b) find the charge on each capacitor when the entire combination is connected across a 6.0-V battery.
- In Problems 69–74, you have three 3.0- μ F capacitors.
69. ■ Find the equivalent capacitance if the three are in parallel.
70. ■■ The three capacitors in parallel are connected across a 30-V battery. Find the charge on each capacitor and potential difference across each.
71. ■ Find the equivalent capacitance if the three capacitors are in series.
72. ■■ The three capacitors in series are connected across a 30-V battery. Find the charge on each capacitor and the potential difference across each.
73. ■■ Two of the capacitors are in parallel, and this parallel combination is in series with the third capacitor. Find the equivalent capacitance.
74. ■■ The capacitor combination in the preceding problem is connected across a 30-V battery. Find the charge on each capacitor and the potential difference across each.
75. ■■ Three capacitors are in series: $C_1 = 45$ μ F, $C_2 = 65$ μ F, and $C_3 = 80$ μ F. (a) Compute the equivalent capacitance. The series combination is connected across a 48-V battery. (b) Find the charge on each capacitor. (c) Compute the energy stored in each capacitor. (d) Show that the total energy you found in part (c) is the same as would be stored in a single capacitor with the equivalent capacitance you found in part (a).
76. ■■ Repeat the preceding problem if the three capacitors are in parallel.
77. **BIO ■■ Defibrillator: energy and power.** A defibrillator uses a 225- μ F capacitor charged to 2400 V. (a) How much energy is stored in the capacitor? (b) What's the power delivered to the patient if the capacitor discharges completely in 2.5 ms?
78. **BIO ■■ Defibrillator: capacitance and energy.** (a) If a 2200-V defibrillator is designed to carry a maximum charge of 0.40 C, what should its capacitance be? (b) How much energy will it store?
79. ■■ A parallel-plate capacitor has plate area 2.5×10^{-3} m² and plate spacing 0.050 mm. (a) What's its capacitance? (b) Find the stored charge and energy when this capacitor is connected across a 50-V battery.
80. ■■ (a) For the preceding problem, find the energy per unit volume between the plates; this is the *electric energy density*. (b) Show that the energy density is equal to $\epsilon_0 E^2/2$, where E is the uniform electric field between the plates. (The expression $\epsilon_0 E^2/2$ is actually the energy density in any electric field, not just this configuration.)
81. ■■ You have a 250-nF capacitor and wish to combine it with one other to make a combined capacitance of 1.50 μ F. How large a capacitor do you need, and should it be combined in series or parallel with the first one?
82. ■■■ You have three capacitors: 10 μ F, 15 μ F, and 20 μ F. Find the values of all the possible capacitances you can create with different combinations using one, two, or all three capacitors.

Section 16.5 Dielectrics

83. ■■ A parallel-plate capacitor has plates with area $A = 3.25 \times 10^{-5}$ m² separated by 0.12 mm. With paper insulation between the plates, find (a) the capacitance; (b) the charge on each plate when a potential difference of 12.0 V is across the capacitor; and (c) the maximum possible potential difference across the plates.
84. ■■ An oil-filled capacitor has square parallel plates 25 cm on a side, separated by 0.50 mm. When the capacitor is charged to 90 V, the stored charge is 7.7 μ C. Find the dielectric constant of the oil.

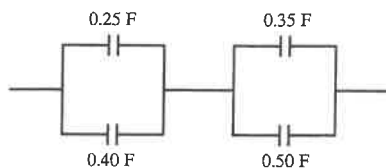


FIGURE P16.68

85. ■■ A 0.45-nF parallel-plate capacitor has plate area $A = 1.7 \times 10^{-4}\text{ m}^2$. If its dielectric has $\kappa = 3.8$, what's the plate separation?
86. ■ The electric field near Earth's surface on a clear day is typically about 100 V/m . How does this compare to the electric field needed for an electrical discharge in air?
87. ■■ A polystyrene-insulated capacitor has $C = 220\text{ nF}$. If the separation between its parallel plates is 0.45 mm , what's the maximum charge it can hold?
88. ■■ The parallel-plate capacitor used on each key of a computer keypad is a square 0.8 cm on a side. Between the plates is a dielectric with $\kappa = 4.2$. When the key is in the "up" position, the plate separation is 0.75 mm . (a) Find the initial capacitance. (b) How far must the key be depressed for a keystroke to be registered, which happens when the capacitance increases by 4.0 pF ?

General Problems

89. ■■ Consider a *spherical capacitor*, in which a spherical conductor of radius a lies inside a concentric spherical conducting shell of radius b . Suppose the inner and outer shells carry charges $\pm Q$, respectively. (a) Compute the potential difference between the two shells. (Note that the electric potential outside of a uniform sphere of charge is the same as if all the charge were concentrated at the sphere's center. Your answer should be a function of Q and the two radii.) (b) Use your answer from part (a) and the definition of capacitance to show that the capacitance of the spherical capacitor is $C = 4\pi\epsilon_0 ab/(b - a)$. (c) Compute the capacitance of a spherical capacitor with inner radius 2.0 cm and outer radius 10.0 cm .
90. ■■■ Consider the spherical capacitor of the preceding problem. Show that in the limit as b approaches a , the capacitance formula is equivalent to that for a parallel-plate capacitor.
91. ■■ A $0.01\text{-}\mu\text{F}$, 300-V capacitor costs $\$0.25$; a $0.10\text{-}\mu\text{F}$, 100-V capacitor costs $\$0.35$; and $30\text{-}\mu\text{F}$, 5.0-V capacitor costs $\$0.88$. (a) Which capacitor can store the most charge? (b) Which can store the most energy? (c) Which is the most effective as an energy storage device, in terms of energy stored per unit cost?
92. ■■ A camera flashtube requires 5.0 J of energy per flash. The flash duration is 1.0 ms . (a) What power does the flashtube use while it's flashing? (b) If the flashtube operates at 200 V , what capacitance is needed to supply the flash energy? (c) If the flashtube is fired once every 10 s , what's the average power consumption?
93. ■■ The NOVA laser fusion experiment at Lawrence Livermore Laboratory in California uses energy at the rate of 10^{14} W (roughly 100 times the output of all the world's power plants) while its lasers are on. However, one laser pulse lasts for only 10^{-9} s . (a) Only about 0.17% of the capacitor energy actually appears as light. How much light energy is delivered in one pulse? (b) The capacitor bank supplying this energy has a total capacitance of 0.26 F . What's the potential difference across the capacitor bank when it's fully charged?
94. ■■ (a) Find the equivalent capacitance of the four identical capacitors shown in Figure GP16.94, measured between points A and B, if $C = 10\text{ }\mu\text{F}$. (b) If a 50-V battery is connected across A and B, what's the charge on each capacitor?

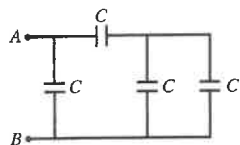


FIGURE GP16.94

95. ■■■ Figure GP16.95 shows something we've heretofore ignored: that there's a nonuniform electric field *outside* a capacitor. Consider the dipole shown in the figure, and show that it experiences an attractive force toward the capacitor. That's the origin of the work done *on* a dielectric slab while it's being inserted into the capacitor, and that's why the stored potential energy is lower in the presence of a dielectric.

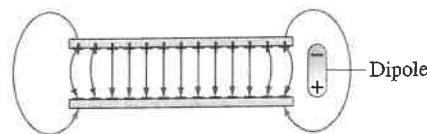


FIGURE GP16.95

96. BIO ■■ **Cell membranes.** The first accurate estimate of the thickness of biological cell membranes used a capacitive technique, in which the capacitance per unit area of cell membrane was determined through a macroscopic measurement of the electric properties of a suspension of cells. The result was a value of about 10^{-2} F/m^2 for a wide range of cells. Assuming a dielectric constant of about 3 for the membrane material, find the membrane thickness. *Note:* Your answer is the thickness of a bipolar lipid layer alone, and is lower by a factor of about 3 than values based on x-ray techniques. The full membrane may be thicker still.
97. ■■■ An unknown capacitance C is in series with a $3.0\text{-}\mu\text{F}$ capacitor. This pair is in parallel with a $1.0\text{-}\mu\text{F}$ capacitor, and the resulting combination is in series with a $2.0\text{-}\mu\text{F}$ capacitor. (a) Make a circuit diagram of this network. (b) When a potential difference of 100 V is applied across the ends of the network, the total energy stored in all capacitors is 5.8 mJ . Find C .
98. BIO ■■ **Neuromuscular microstimulator.** A neuromuscular microstimulator is a device designed to be implanted into paralyzed muscles or other tissue suffering neuromuscular dysfunction. The device supplies an electric current that stimulates the dysfunctional tissue. A particular device uses a capacitor built into a microchip; the capacitor can supply 10 mA of current for $200\text{ }\mu\text{s}$ before it's fully discharged. (a) Find the initial charge on the capacitor and (b) the capacitance if it's initially charged to 4.5 V .

Answers to Chapter Questions

Answer to Chapter-Opening Question

The energy is stored in the electric field of a pair of conductors—a capacitor—and dumped quickly through the defibrillator into the patient's chest when needed.

Answers to GOT IT? Questions

- Section 16.1 (c) 0.27 J
 Section 16.2 (d) $1.6 \times 10^{-17}\text{ J}$
 Section 16.3 (d) 3.0 MV/m
 Section 16.4 (b) $>$ (d) $>$ (c) $>$ (a)